

AP CALCULUS AB - UNIT 1

1.13 Removing Discontinuities

Lesson Overview

- Decide whether a discontinuity can be removed by changing a point value.
- Find parameters that make piecewise functions continuous at joins.
- Create or cancel common factors to identify removable holes.
- Solve multi-boundary continuity systems and justify uniqueness or impossibility.

Key Knowledge and Vocabulary

Repair value. If $\lim_{x \rightarrow c} f(x) = L$ is finite, define or redefine $f(c) = L$.

Join condition. At a piecewise boundary require left limit = right limit = assigned value.

Factor creation. A parameter may be chosen so the numerator shares a denominator factor.

Multiplicity. Cancellation repairs a hole only when every problematic denominator factor is removed.

Possible outcomes. A repair problem can have a unique solution, infinitely many solutions, no solution, or require changing more than a point value.

Explanation and Strategy

Step 1. First classify the discontinuity. A point-value repair is possible only when a finite common limit already exists.

Step 2. For piecewise joins, write equations before solving parameters.

Step 3. For rational expressions, use the factor theorem: $x - c$ divides a polynomial $P(x)$ exactly when $P(c) = 0$.

Solved Examples

Worked Example: Fill a hole

Define $f(3)$ so that $f(x) = \frac{x^2-9}{x-3}$ is continuous at 3.

Solution. For $x \neq 3$, $f(x) = x + 3$, so the limit is 6. Define $f(3) = 6$.

Worked Example: One parameter at a join

Find k so $f(x) = kx + 1$ for $x < 2$ and $f(x) = x^2 - 1$ for $x \geq 2$ is continuous at 2.

Solution. Require $2k + 1 = 3$, giving $k = 1$.

Worked Example: Two joins

Find a, b so $f(x) = x + 1$ for $x < 0$, $f(x) = ax + b$ for $0 \leq x < 2$, and $f(x) = 5$ for $x \geq 2$ is continuous.

Solution. At 0, $b = 1$. At 2, $2a + b = 5$, so $a = 2$. Thus $(a, b) = (2, 1)$.

Leveled Practice

Easy Level

Foundational fluency. Focus on notation, definitions, and direct procedures.

1. What value repairs $f(x) = \frac{x^2-4}{x-2}$ at $x = 2$?

2. What value repairs $g(x) = \frac{x^2+x-6}{x-2}$ at $x = 2$?

3. Can changing one point repair a jump discontinuity?

4. Can changing one point repair an infinite discontinuity?

Medium Level

Apply the idea in one or two connected steps.

5. Find k so $f(x) = \begin{cases} 2x + k, & x < 1 \\ x^2 + 3, & x \geq 1 \end{cases}$ is continuous at 1.

6. Find a so $\frac{x^2+ax-6}{x-2}$ has a removable discontinuity at 2. Then find the repair value.

7. Define $h(0)$ so $h(x) = x \sin(1/x)$ for $x \neq 0$ is continuous.

8. Find k so $f(x) = \begin{cases} \sqrt{x+3}, & x < 1 \\ k, & x = 1 \\ 3-x, & x > 1 \end{cases}$ is continuous at 1.

Hard Level

Connect representations, conditions, and justification.

9. Find a, b so $f(x) = \begin{cases} ax + b, & x < 1 \\ x^2, & 1 \leq x < 3 \\ 2x + b, & x \geq 3 \end{cases}$ is continuous at both joins.

10. For what m does $\frac{x^2+mx+16}{x+4}$ have a removable discontinuity at -4 ? Find the hole height.

11. Find k so $f(x) = \frac{x^2-k^2}{x-k}$ for $x \neq k$ and $f(k) = 12$ is continuous at k .

12. Explain why no choice of k makes $f(x) = 0$ for $x < 1$, $f(1) = k$, and $f(x) = 2$ for $x > 1$ continuous.

Difficult Level

Use nonroutine structure and precise mathematical reasoning.

13. Find all a such that $\frac{x^2 + (a-3)x - 3a}{(x-3)^2}$ has a removable discontinuity at 3.

14. Find a, b so $P(x) = x^3 + ax + b$ is divisible by $(x - 2)^2$.

15. A piecewise continuity system reduces to $a + b = 2$, $2a + 2b = 5$. Diagnose the outcome.

Challenge Level

Synthesize, construct, generalize, or prove.

16. For positive integer n , choose coefficients so $\frac{x^n - a^n}{x - a}$ extends continuously to $x = a$, and give the extension value.

17. Construct a one-parameter family of distinct functions, each with a removable discontinuity at $x = 2$ and repair value 7.

AP-Style Practice

Multiple-Choice Practice – No Calculator

Choose the best answer. Use exact values unless a decimal approximation is requested.

- To repair a removable discontinuity at $x = c$, define $f(c)$ to be
 - 0
 - the left-hand limit only
 - the finite two-sided limit
 - any real number
- For $\frac{x^2+kx-12}{x-3}$ to have a removable discontinuity at 3, k must be
 - 7
 - 1
 - 1
 - 7
- Which type cannot be repaired by changing only $f(c)$?
 - A hole with finite limit
 - A wrong point value
 - A jump
 - An undefined point with finite common limit

Multiple-Choice Practice – Calculator Required

Choose the best answer. Use exact values unless a decimal approximation is requested.

- A calculator estimates a common limit of 4.999999 at $x = 2$ while $f(2)$ is undefined. The natural exact repair conjecture is
 - 0
 - 2
 - 5
 - $+\infty$
- For $f(x) = \frac{x^3-8}{x-2}$, a graphing utility hides the hole. What value should fill it?
 - 4
 - 8
 - 12
 - 16

Free-Response Practice – No Calculator

$$\text{Let } f(x) = \begin{cases} x + 4, & x < 1 \\ ax + b, & 1 \leq x < 3. \\ x^2 - 1, & x \geq 3 \end{cases}$$

(a) Write the two continuity equations.

[2 points]

(b) Find a and b .

[2 points]

(c) Justify that these values make f continuous on all real numbers.

[2 points]

Free-Response Practice – Calculator Required

A calculator suggests that $R(x) = \frac{x^3 + ax^2 + bx - 6}{x - 1}$ may have a removable discontinuity at 1 and a hole value of 4.

(a) Use the removable condition to obtain one equation in a, b .

[2 points]

(b) Use the desired hole value to obtain a second equation.

[3 points]

(c) Find a, b .

[2 points]